

Large language model-powered meta-analysis of mobile interventions for stress management

By leveraging large language models and Bayesian network meta-analysis of 63 randomized controlled trials, this study developed a three-dimensional classification of stress management interventions delivered with mobile technology. This analysis review found no evidence that human or technological support enhances the effects of the interventions, and it offers scalable strategies to reduce mental health disparities.

This is a summary of:

Zhu, H. et al. A systematic review and Bayesian network meta-analysis on the efficacy and potential of mobile interventions for stress management. *Nat. Hum. Behav.* <https://doi.org/10.1038/s41562-025-02162-0> (2025).

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Published online: 29 April 2025

The problem

Stress has become a pervasive global health challenge. Traditional in-person mental health interventions (such as therapy sessions) have shown good effectiveness but can be limited by cost, location and scheduling constraints. By contrast, digital mental health interventions (for example, via mobile applications or web tools) have proliferated rapidly, yet their efficacy, accessibility and equity remain uncertain. Existing meta-analyses of studies that assess mobile technology for mental health interventions often prioritize anxiety and depression outcomes, which leaves stress-specific interventions underexplored. Comparisons between studies are further complicated by heterogeneity in intervention designs – such as varying theoretical foundations (evidence-based psychological theories that target different factors), levels of human support (additional assistance from professional mental healthcare workers) and degrees of mobile technology use (complexity of technology). Crucial gaps persist that include unclear classification systems to disentangle the effects of different intervention components; insufficient evidence on how moderators (for example, human support and technology use) affect outcomes; and limited scalability insights for low-resource settings. Addressing these gaps is vital to advancing equitable mental healthcare and aligning with the United Nations Sustainable Development Goals (SDGs)¹.

The solution

By integrating expert insights with large language models (LLMs; for example, ChatGPT)^{2,3}, this study introduces a three-dimensional framework (theoretical foundation, human support and mobile technology) to systematically classify mobile-technology-based stress interventions (Fig. 1a). We searched eight databases from their inception to 20 November 2023, and we identified studies published as early as 1980. A total of 63 randomized controlled trials on internet-based interventions for stress reduction were included, which covered over 20,000 participants from healthy populations exposed to high stress levels (such as emergency lines telecommunicators) and adults with subclinical symptoms who do not meet clinical diagnostic criteria. Guided by the three-dimensional framework, we identified 19 types of mobile intervention for stress management.

We used ChatGPT-4 to expedite literature screening and coding with high consistency. The consistency index for literature extraction between the first author and ChatGPT-4 was 94.2%. Additionally, the reliability of coding

consistency between ChatGPT-4's classification of framework parameters and human coders' classification was 0.78, which demonstrates the potential of LLMs to streamline systematic reviews. Leveraging Bayesian network meta-analysis and Bayesian meta-regression, we identified top-performing interventions. Bayesian network meta-analysis is a powerful evidence-synthesis method that simultaneously evaluates multiple treatments and their indirect relationships. It integrates prior knowledge to strengthen conclusions, reduces the need for additional direct comparative studies, and minimizes potential biases from random error more effectively than standard meta-analysis. Stress management programmes, problem-solving therapy and mindfulness meditation ranked highest in efficacy, even without human support or advanced technology (Fig. 1b). Contrary to previous studies on the roles of human support and technology^{4,5}, neither significantly enhanced the outcomes in our analysis. Additionally, age moderated the efficacy of the interventions, which indicates that younger participants experienced greater stress-reduction effects.

The implications

This work provides a multi-dimensional classification system for mobile interventions for stress management and offers actionable insights for policymakers and developers of stress management interventions. By identifying cost-effective strategies (for example, self-administered tools driven by psychological theories) and challenging assumptions about the need for human or technological support, it paves the way for equitable, evidence-based mental health solutions. The integration of LLMs into research workflows also sets a precedent for efficient, reproducible meta-analyses, to accelerate innovation in the digital era.

Despite our three-dimensional framework and comprehensive meta-analysis, the study is limited by its exclusive focus on mobile interventions that reduce stress, which restricts generalization to other mental health conditions (such as depression). Future research should extend the classification system to other mental health domains and refine its dimensions (for example, introducing gradations of human support) for more-detailed insights. Advancing LLMs and machine learning can help to develop adaptive interventions that dynamically respond to user needs, and improve engagement and efficacy. Prioritizing scalable, low-cost solutions in underserved regions remains a crucial next step⁴.

Yiqun Gan & Jinmeng Liu

Peking University, Beijing, China.

EXPERT OPINION

"This is a generally well-conducted meta-analysis examining the comparative efficacy of various mobile interventions for stress management. The authors used standard meta-analytic methods and thoughtfully addressed the risk of bias. I appreciate their efforts in pre-registering the study and in proposing a conceptual framework for

their findings. I also commend the authors for choosing the Bayesian network meta-analysis approach over the traditional meta-analysis method to explore the comparative efficacy of different self-administrated mobile mental health interventions."

Sin U Lam, University of Wisconsin–Madison, Madison, WI, USA.

FIGURE

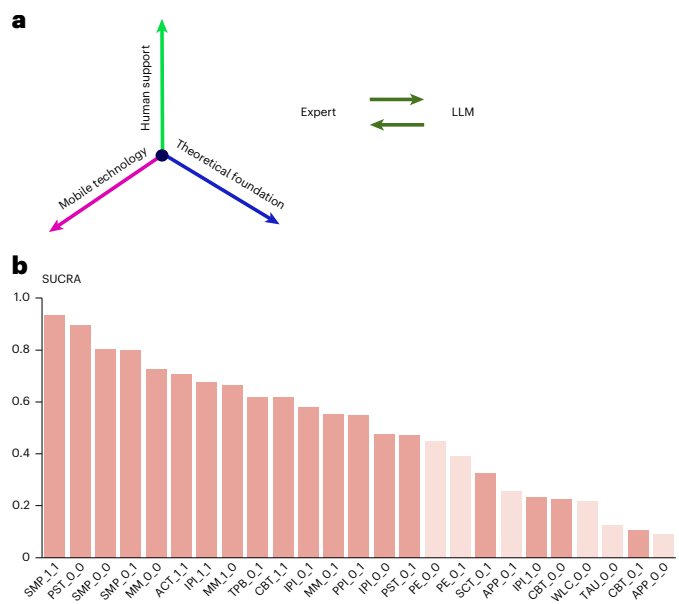


Fig. 1 | Classification framework and ranking for mobile stress management interventions. a, This framework classifies mobile interventions across three dimensions: human support, mobile technology and theoretical foundation. b, The surface under the cumulative ranking (SUCRA) indicates the likelihood of each intervention being one of the most effective: a value of 1 indicates the highest probability of being the best option. Lighter shades indicate the control groups. ACT, acceptance and commitment therapy; APP, attention or psychological placebo; CBT, cognitive behavioural therapy; IPI, integrated psychotherapy intervention; MM, mindfulness and meditation; PE, psychoeducation; PPI, positive psychology intervention; PST, problem-solving therapy; SCT, social cognitive theory; SMP, stress management programme; TAU, treatment as usual; TPB, theory of planned behaviour; WLC, waiting list control. The first number after each intervention indicates human support; the second, mobile technology; 1, yes; 0, none. © 2025, Zhu, H. et al.

BEHIND THE PAPER

Amid the COVID-19 pandemic, we developed online stress-reduction programmes for affected populations. Although expert-guided interventions showed marginally better outcomes than self-help approaches in early trials, disparities in access persisted across remote regions. To resolve uncertainties, we conducted a meta-analysis, updated twice to capture new papers. Overwhelmed by the workload, we innovated by using ChatGPT for literature screening and coding, which achieved almost complete agreement with results by human coders. Surprisingly, Bayesian meta-regression analysis revealed

that neither human support nor mobile technology significantly boosted efficacy – contradicting common assumptions. Equally unexpectedly, LLMs streamlined intervention classification, which improved scalability. These early attempts make it possible to conduct LLM-powered meta-analyses. Our findings advocate for inclusive digital mental health solutions that bypass traditional barriers, which offers hope for underserved communities to bridge care gaps without relying on extensive infrastructure or specialized staff.

Y.G. & J.L.

REFERENCES

1. WHO. *World Health Statistics 2020: Monitoring Health for the SDGs, Sustainable Development Goals* (WHO, 2020).
A report on the latest global health statistics for tracking progress toward the UN SDGs.
2. Wernitz, A., Amado, S., Jasman, M., Ervin, A. & Rhodes, J. E. Providing human support for the use of digital mental health interventions: systematic meta-review. *J. Med. Internet Res.* **25**, e42864 (2023).
A systematic meta-review that evaluates the role of human support in improving the effectiveness of digital mental health interventions.
3. Demszky, D. et al. Using large language models in psychology. *Nat. Rev. Psychol.* **2**, 688–701 (2023).
A review that explores the applications, challenges and ethical considerations of using LLMs in psychological research.
4. Cookson, R. et al. Using cost-effectiveness analysis to address health equity concerns. *Value Health* **20**, 206–212 (2017).
An empirical study that demonstrates how cost-effectiveness analysis can be used to address health equity concerns in resource allocation decisions.
5. Wernitz, A. et al. Providing human support for the use of digital mental health interventions: systematic meta-review. *J. Med. Internet Res.* **25**, e42864 (2023).
A systematic meta-review that highlights the potential of human support in improving the effects of digital mental health interventions.

FROM THE EDITOR

"As the prevalence of stress is increasing, effective stress management is becoming crucial — but which approach is the best? This systematic review and network meta-analysis of 63 randomized controlled trials suggests that stress management programmes, problem-solving therapy and mindfulness meditation are most effective in managing stress." **Editorial Team, Nature Human Behaviour.**